

SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

Faculty of Computing — Department of Software Engineering

Research Project — Periodic Log Entry Form

Project R26-SE-009 | Agent 3 | Hamna Hakeem (IT22516916) | Reporting period 23.03.2026 – 25.08.2026

1. PROJECT AND STUDENT DETAILS

Project ID	R26-SE-009
Research Topic	Self-Optimized LLM Multi-Agent System for Software Engineering
Supervisor	Prof. Nuwan Kodagoda
Co-Supervisor	Mr. Eishan Weerasinghe
Student Name	Hamna Hakeem
Student Registration No.	IT22516916
Individual Component	Agent 3 — QA Tester (AI QA Engineer) and the QA user interface
Reporting Period	23 March 2026 to 25 August 2026
Phase Covered	Requirement, implementation and integration phase, representing approximately 65% of the planned project work.

2. SUMMARY OF RP WORK CARRIED OUT DURING THE REPORTING PERIOD

During this reporting period Agent 3 was developed as the quality assurance and validation component of the multi-agent system, together with the QA-facing user interface. The early part of the period structured the QA workflow: unit testing, integration testing, API contract validation, quality scoring, issue reporting and round-based feedback to Agent 2 for corrective action, with PDF report output. The middle of the period delivered the QA front end — the cycle timeline, bug reporting and contract review screens, evidence and artefact views, memory logs and project history — together with the Node.js analyser and dependency graph used when preparing generated projects for testing. The later part of the period covered the redesign of the QA backend into staged unit, runtime/API, end-to-end and security verification, with an installable test harness, mock handling and isolated per-run sessions. Group contributions were made throughout so that testing progress, bug visibility and the QA gate before deployment are demonstrable inside the shared workspace.

3. DATE-WISE RP WORK (40 ENTRIES, VERIFIED MONTHLY)

MARCH 2026		5 ENTRIES	
NO.	DATE	RP WORK CARRIED OUT	
1	23.03.2026	Defined the scope of Agent 3 as the AI QA Tester responsible for validating generated systems and returning actionable defects to the builder agent.	
2	24.03.2026	Designed the QA workflow to cover unit testing, integration testing, API contract checks, scoring and bug reporting across the multi-agent pipeline.	
3	26.03.2026	Identified the core test stages and reporting outputs needed for a repeatable quality gate before deployment.	
4	28.03.2026	Planned issue classification with severity levels, module mapping, fix recommendations and round-based re-testing behaviour.	
5	31.03.2026	Defined the quality score structure and summary metrics required for measuring functional quality, security, performance and requirement coverage.	
MONTHLY VERIFICATION — MARCH 2026			
SUPERVISOR'S COMMENTS FOR MARCH 2026		SUPERVISOR'S SIGNATURE	CO-SUPERVISOR'S SIGNATURE
		Signature & Date	Signature & Date

APRIL 2026		10 ENTRIES	
NO.	DATE	RP WORK CARRIED OUT	
6	02.04.2026	Designed the feedback loop from Agent 3 back to Agent 2 so failed runs could trigger guided repair instead of manual debugging only.	
7	04.04.2026	Planned report artefact generation for QA summaries and issue bundles that could be consumed by the rest of the team.	
8	07.04.2026	Refined the staged testing cycle so validation progress, fixes and final pass state could be presented clearly to users.	
9	09.04.2026	Coordinated with the group workflow so QA status could act as the deployment gate for Agent 4 and appear in the shared dashboard.	
10	21.04.2026	Implemented the unit-testing and integration-testing routines together with the PDF report writer used for QA output.	
11	23.04.2026	Implemented the Agent 3 backend in code, including test-run start/status endpoints, progressive logs, summary metrics, issue lists and JSON report output.	
12	24.04.2026	Built the shared QA Center interface with a visual test timeline, bug report pane, fix log animation, unit/integration/API tabs and QA inspector.	
13	24.04.2026	Added round-based bug tracking and simulated repair handoff behaviour so Agent 3 could demonstrate interaction with Agent 2 inside the group workspace.	
14	27.04.2026	Reviewed the shared dashboard and activity integration so QA outcomes, open bugs and handoff status are visible across the group demo.	
15	28.04.2026	Prepared the reporting-period summary and verified Agent 3 alignment with the first-phase project milestone.	
MONTHLY VERIFICATION — APRIL 2026			
SUPERVISOR'S COMMENTS FOR APRIL 2026		SUPERVISOR'S SIGNATURE	CO-SUPERVISOR'S SIGNATURE
		Signature & Date	Signature & Date

MAY 2026		11 ENTRIES
NO.	DATE	RP WORK CARRIED OUT
16	04.05.2026	Set up the QA front-end routes and the base screen structure for the testing workspace.
17	05.05.2026	Built the bug-reporting and contract-review screens used to present QA findings.
18	05.05.2026	Updated the bug-reporting and contract-review screens following the first internal review round.
19	05.05.2026	Implemented the QA cycle timeline and the testing progress states.
20	06.05.2026	Added the memory-log and project-history views so previous QA runs remain visible to the user.
21	07.05.2026	Built the artefacts and evidence presentation screens for test output.
22	09.05.2026	Added the shared QA components and wired the QA routes into the common navigation.
23	10.05.2026	Completed responsive checks and polish across the QA screens.
24	10.05.2026	Cleaned the front-end working branch and prepared it for packaging.
25	11.05.2026	Added the Node.js analyser, generator and dependency graph support used when preparing generated projects for testing.
26	11.05.2026	Finalised the Agent 3 front-end copy and pushed the reviewed build to the shared repository.

MONTHLY VERIFICATION — MAY 2026		
SUPERVISOR'S COMMENTS FOR MAY 2026	SUPERVISOR'S SIGNATURE	CO-SUPERVISOR'S SIGNATURE
	Signature & Date	Signature & Date

JUNE 2026		4 ENTRIES
NO.	DATE	RP WORK CARRIED OUT
27	08.06.2026	Reviewed the version 1 QA cycle results and identified where staged demonstration testing had to be replaced by real test execution.
28	15.06.2026	Studied test-harness approaches for generated JavaScript and React projects and compared the available runner options.
29	22.06.2026	Planned the redesigned QA stages covering unit, runtime/API, end-to-end and security verification.
30	29.06.2026	Defined the evidence model so that every QA verdict is backed by a stored artefact rather than a log line.

MONTHLY VERIFICATION — JUNE 2026		
SUPERVISOR'S COMMENTS FOR JUNE 2026	SUPERVISOR'S SIGNATURE	CO-SUPERVISOR'S SIGNATURE
	Signature & Date	Signature & Date

JULY 2026			4 ENTRIES		
NO.	DATE	RP WORK CARRIED OUT			
31	06.07.2026	Specified the unit-test authoring and repair loop so failing generated tests are corrected instead of discarded.			
32	13.07.2026	Designed the end-to-end stage: preflight checks, console capture, normalisation and grounding of observed behaviour.			
33	20.07.2026	Prepared the API-verification and security-check requirements for the QA backend rebuild.			
34	27.07.2026	Reviewed the integration points between the QA agent and the builder so failures return actionable repair context.			
MONTHLY VERIFICATION — JULY 2026					
SUPERVISOR'S COMMENTS FOR JULY 2026			SUPERVISOR'S SIGNATURE		CO-SUPERVISOR'S SIGNATURE
			Signature & Date		Signature & Date

AUGUST 2026			6 ENTRIES		
NO.	DATE	RP WORK CARRIED OUT			
35	16.08.2026	Added the QA agent component and the testing user interface into the shared Studio.			
36	18.08.2026	Aligned the QA run API with the shared runs service so test progress can be polled from the interface.			
37	22.08.2026	Reviewed the tester module split and the security shim added on the builder side for the QA interface.			
38	24.08.2026	Split the test-harness foundations into reusable modules.			
39	25.08.2026	Added harness installation support and mock handling so test runs execute in isolation.			
40	25.08.2026	Split the QA session foundations so each test run keeps its own isolated state and snapshot.			
MONTHLY VERIFICATION — AUGUST 2026					
SUPERVISOR'S COMMENTS FOR AUGUST 2026			SUPERVISOR'S SIGNATURE		CO-SUPERVISOR'S SIGNATURE
			Signature & Date		Signature & Date

4. CONSOLIDATED MONTHLY VERIFICATION

MONTH	ENTRIES LOGGED	SUPERVISOR'S SIGNATURE	DATE
March 2026	5		
April 2026	10		
May 2026	11		
June 2026	4		
July 2026	4		
August 2026	6		
Total	40		

5. PROJECT WORK PLANNED BEFORE THE NEXT EVALUATION PERIOD

1. Complete the end-to-end stage — preflight, console capture, normalisation and grounding — and converge it against real generated applications.
2. Finish the unit-test authoring and repair loop so failing generated tests are corrected rather than discarded.
3. Complete API verification and the security checks, and bind every verdict to stored evidence.
4. Improve the repair-feedback loop so Agent 2 receives clearer defect context and regression indicators.
5. Tighten the QA handoff artefacts consumed by Agent 4 so deployment gating is stricter and more reliable.

6. DECLARATION AND SIGNATURES

I certify that the work recorded in this log book for the period 23 March 2026 to 25 August 2026 was carried out by me as the individual component contribution to research project R26-SE-009, and that the group-level contributions recorded here were made jointly with the other project members.

STUDENT Hamna Hakeem (IT22516916) — Signature & Date	SUPERVISOR Prof. Nuwan Kodagoda — Signature & Date
CO-SUPERVISOR Mr. Eishan Weerasinghe — Signature & Date	OVERALL SUPERVISOR COMMENTS Comments